

# Mathematical Foundations of Machine Learning

## Week 6: Schur Decomposition

Ayberk Zeytin

1. Recap : Diagonalization, Spectral theorem
2. Schur decomposition

### 1 Quick Recap — Week 5

#### 1.1 Eigenvalues of Special Transformations

**Theorem 1.1.** *If  $T : V \rightarrow V$  is represented in some basis by an upper (or lower) triangular matrix, then its eigenvalues are precisely the diagonal entries of that matrix. Equivalently: the eigenvalues of  $T$  are the diagonal entries of  $[T]_{\mathcal{B}}$  for any basis in which  $T$  is triangular.*

**Example** (Triangular case). Let  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be

$$T(x, y, z) = (x + 2y - 2z, 3y + 5z, -z).$$

Relative to the standard basis, the matrix is  $\begin{bmatrix} 1 & 2 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{bmatrix}$  (upper triangular). Hence the eigenvalues are 1, 3, -1.

#### 1.2 Diagonalization as a Coordinate Simplification

**Definition.** A linear map  $T : V \rightarrow V$  is **diagonalizable** if  $V$  has a basis of eigenvectors of  $T$ . In that basis,  $[T]_{\mathcal{B}}$  is diagonal with the eigenvalues on the diagonal.

**Example.**  $T(x, y) = (x + y, 2y)$  has eigenvalues  $\lambda = 1, 2$  with eigenvectors  $(1, 0)$  and  $(1, 1)$ . In the basis  $\{(1, 0), (1, 1)\}$ :

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

**Definition** (Multiplicities). Given a transformation  $T : V \rightarrow V$  we define :

- **Algebraic Multiplicity (AM):** Multiplicity of  $\lambda$  as a root of  $p_A(\lambda)$ .
- **Geometric Multiplicity (GM):** Dimension of the eigenspace corresponding to the eigenvalue  $\lambda$ , i.e.  $\dim(\{\mathbf{v} \in V \mid T\mathbf{v} = \lambda\mathbf{v}\}) = \dim(\ker(A - \lambda I))$ .

**Example** (Non-Diagonalizable Case: Shear Map Revisited). The shear map matrix  $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  has characteristic polynomial  $p_A(\lambda) = (1 - \lambda)^2$ , so  $\lambda = 1$  is the only eigenvalue, with **AM**(1) = 2.

The eigenspace is  $\ker(A - I) = \ker \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ , which spans  $\text{span}\{(1, 0)\}$ . Thus, **GM**(1) = 1. Since  $1 = \mathbf{GM} < \mathbf{AM} = 2$ , the matrix  $A$  is **not diagonalizable**. It is a prototypical example of a non-diagonalizable matrix that requires the Jordan Canonical Form for simplification.

**Example** (Non-diagonalizable (why GM < AM matters)).

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad p_A(\lambda) = (1 - \lambda)^3, \quad \dim \ker(A - I) = 1.$$

So  $A$  is not diagonalizable: the eigenvectors do not span  $V$ .

**Theorem 1.2** (Diagonalizability criterion).  $T$  is diagonalizable iff  $V$  has a basis of eigenvectors of  $T$ . Equivalently (over a field where  $p_T$  splits),  $\mathbf{GM}(\lambda) = \mathbf{AM}(\lambda)$  for every eigenvalue  $\lambda$ .

*Summary.* Eigenvectors/eigenvalues are basis-free descriptors of  $T$ . In an eigenbasis,  $T$  becomes scaling along invariant directions, greatly simplifying analysis, geometry and computation.

**Example.** For  $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ,

$$\det(A - \lambda I) = (1 - \lambda)(2 - \lambda)(3 - \lambda).$$

Distinct eigenvalues  $\Rightarrow$  three linearly independent eigenvectors  $\Rightarrow$  diagonalizable. One finds

$$D = \text{diag}(1, 2, 3), \quad P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A = PDP^{-1}.$$

**Example.** For  $B = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ,  $\lambda = 2$  has algebraic multiplicity 2 but eigenspace dimension 1 (GM < AM), so  $B$  is not diagonalizable.

### 1.3 Spectral Theorem

**Theorem 1.3** (Real symmetric case). *If  $A = A^\top$ , then  $A = QDQ^\top$  with  $Q$  orthogonal and  $D$  real diagonal. Eigenvalues are real; eigenspaces for distinct eigenvalues are orthogonal.*

**Example.** Consider the real symmetric matrix  $A$ :

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$$

The Spectral Theorem guarantees that  $A$  is orthogonally diagonalizable, meaning its eigenvectors will be orthogonal.

**1. Find Eigenvalues ( $\lambda$ ):** We solve the characteristic equation  $\det(A - \lambda I) = 0$ :

$$\begin{aligned} \det \begin{pmatrix} 4 - \lambda & 1 \\ 1 & 4 - \lambda \end{pmatrix} &= (4 - \lambda)^2 - 1^2 = 0 \\ (4 - \lambda)^2 &= 1 \\ 4 - \lambda &= \pm 1 \end{aligned}$$

This yields two distinct real eigenvalues:

$$\lambda_1 = 4 - 1 = 3 \quad \text{and} \quad \lambda_2 = 4 + 1 = 5$$

**2. Find Eigenvectors ( $v$ ):** ►For  $\lambda_1 = 3$ : Solve  $(A - 3I)v_1 = 0$ .

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x_1 + x_2 = 0 \Rightarrow x_2 = -x_1$$

Choosing  $x_1 = 1$ , we get  $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ .

►For  $\lambda_2 = 5$ : Solve  $(A - 5I)v_2 = 0$ .

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -x_1 + x_2 = 0 \Rightarrow x_1 = x_2$$

Choosing  $x_1 = 1$ , we get  $v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ .

**3. Construct Orthogonal Diagonalization ( $A = PDP^T$ ):** First, we normalize the eigenvectors to form an orthonormal set  $\mathbf{u}_1$  and  $\mathbf{u}_2$ :

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The diagonal matrix  $D$  and the orthogonal matrix  $P$  are:

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \quad \text{and} \quad P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

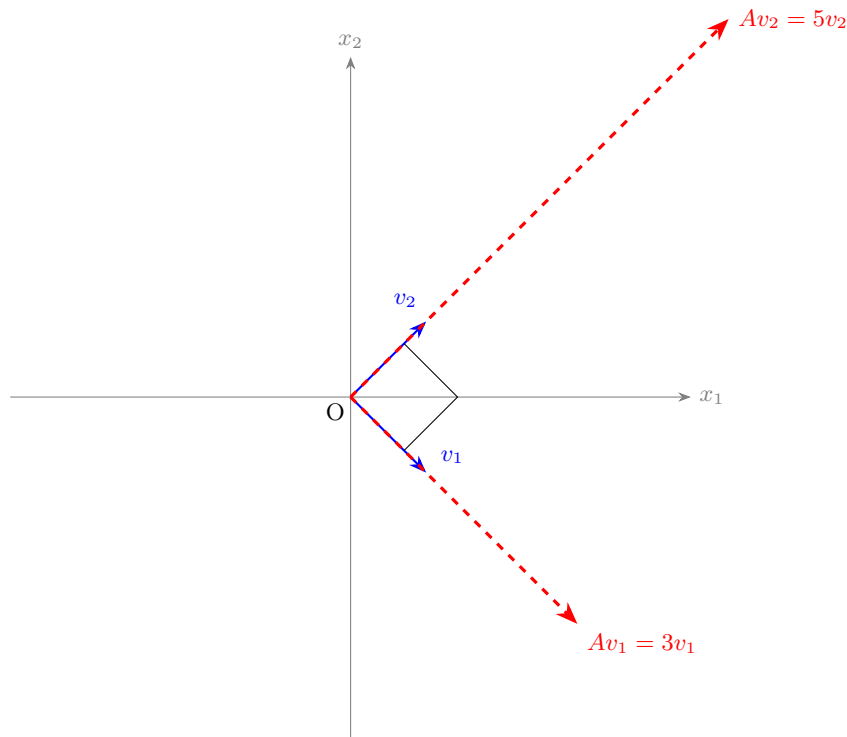
The orthogonal diagonalization is  $A = PDP^T$ .

The transformation  $T(\mathbf{x}) = A\mathbf{x}$  stretches the vector  $v_1$  by a factor of 3 and stretches  $v_2$  by a factor of 5, while preserving their directions.

**Example.** Consider the matrix  $A$ :

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

This matrix is diagonalizable because it has three distinct eigenvalues.



**1. Find Eigenvalues ( $\lambda$ ):** Since  $A$  is a block upper triangular matrix, the determinant  $\det(A - \lambda I)$  is the product of the determinants of the diagonal blocks.

$$\begin{aligned}\det(A - \lambda I) &= \det \begin{pmatrix} 2 - \lambda & 0 & 0 \\ 1 & 2 - \lambda & 1 \\ -1 & 0 & 1 - \lambda \end{pmatrix} \\ &= (2 - \lambda) \det \begin{pmatrix} 2 - \lambda & 1 \\ 0 & 1 - \lambda \end{pmatrix} \\ &= (2 - \lambda)(2 - \lambda)(1 - \lambda) = 0\end{aligned}$$

The eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 2$$

Here,  $\lambda = 2$  has an Algebraic Multiplicity (**AM**) of 2.

**2. Find Eigenvectors ( $v$ ) and Check Diagonalizability:** For  $A$  to be diagonalizable, the Geometric Multiplicity (**GM**) of  $\lambda = 2$  must also be 2.

►For  $\lambda_1 = 1$  (**AM** = 1): Solve  $(A - 1I)v_1 = 0$ .

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & 0 \end{pmatrix} v_1 = 0$$

The resulting system  $x_1 = 0$ ,  $x_1 + x_2 + x_3 = 0$ ,  $-x_1 = 0$  implies  $x_1 = 0$  and  $x_2 = -x_3$ . Setting  $x_3 = 1$ , we get the eigenvector:

$$v_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

Since **AM** = 1, we have **GM** = 1.

►For  $\lambda_{2,3} = 2$  (**AM** = 2): Solve  $(A - 2I)v = 0$ .

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{pmatrix} v = 0$$

The second and third rows are redundant, giving the single equation  $x_1 + x_3 = 0$ , or  $x_1 = -x_3$ . The variable  $x_2$  is free. We have two free variables ( $x_2$  and  $x_3$ ), so **GM**(2) = 2. The eigenvectors are:

$$v = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -x_3 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Choosing two linearly independent eigenvectors:

$$v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Since  $\mathbf{GM}(2) = \mathbf{AM}(2) = 2$ , the matrix  $A$  is diagonalizable.

**3. Construct Diagonalization ( $A = PDP^{-1}$ ):** The diagonal matrix  $D$  and the change-of-basis matrix  $P$  are:

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad \text{and} \quad P = (v_1 \ v_2 \ v_3) = \begin{pmatrix} 0 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

For a  $3 \times 3$  matrix, the visualization is generally harder to draw on a 2D page, but we can illustrate the effect on the three orthogonal eigen-directions defined by the eigenvectors. In this specific example,  $v_2$  and  $v_3$  span a 2-dimensional eigenspace ( $\mathbf{E}_2$ ), which is scaled by  $\lambda = 2$ , and  $v_1$  is scaled by  $\lambda = 1$ .

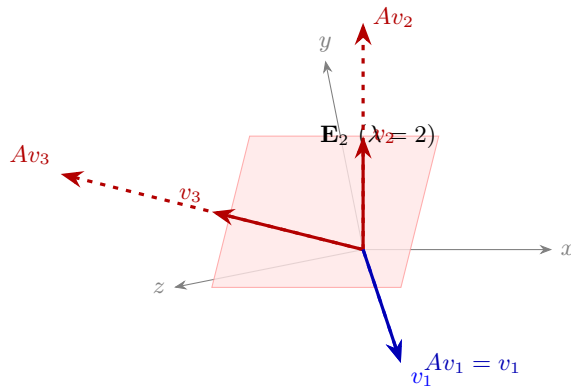


Figure 1: Conceptual visualization of the  $3 \times 3$  transformation  $T(x) = Ax$ . The 2D eigenspace  $\mathbf{E}_2$  is scaled by  $\lambda = 2$ , and the vector  $v_1$  (scaled by  $\lambda = 1$ ) defines the third eigen-direction.

## 2 Jordan Canonical Form: The Non-Diagonalizable Case

Let's first define

**Definition** (Generalized Eigenvector). Let  $A$  be an  $n \times n$  matrix and  $\lambda$  an eigenvalue. A nonzero vector  $\mathbf{v}$  is a **generalized eigenvector of rank  $k$**  corresponding to  $\lambda$  if:

- $(A - \lambda I)^k \mathbf{v} = \mathbf{0}$ , and
- $(A - \lambda I)^{k-1} \mathbf{v} \neq \mathbf{0}$

Notice that, a standard eigenvector is a **generalized eigenvector** of rank 1. These vectors form a sequence known as a **Jordan chain** of length  $k$ , generated by the vector  $\mathbf{v}_k$  (the generalized eigenvector of rank  $k$ ):

$$\mathbf{v}_1 = (A - \lambda I)\mathbf{v}_2, \quad \mathbf{v}_2 = (A - \lambda I)\mathbf{v}_3, \quad \dots, \quad \mathbf{v}_{k-1} = (A - \lambda I)\mathbf{v}_k.$$

Since  $(A - \lambda I)\mathbf{v}_1 = (A - \lambda I)^2\mathbf{v}_2 = \dots = (A - \lambda I)^k\mathbf{v}_k = \mathbf{0}$ , the first vector  $\mathbf{v}_1$  in the chain is a standard eigenvector.

The rank of a generalized eigenvector  $\mathbf{v}$  corresponds to its position in a Jordan chain. The vector of rank  $k$  is the one that **starts** the chain of length  $k$ .

**Example** (Generalized Eigenvector of Rank 2). Consider the  $2 \times 2$  non-diagonalizable matrix

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

with the single eigenvalue  $\lambda = 1$ .

The chain is defined by:

1.  $(B - I)\mathbf{v}_1 = \mathbf{0}$  (Eigenvector, Rank 1)
2.  $(B - I)\mathbf{v}_2 = \mathbf{v}_1$  (Generalized Eigenvector, Rank 2)

**Solution:**

► **Rank 1 ( $\mathbf{v}_1$ ):** Solving  $(B - I) \mathbf{v}_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$  yields the eigenvector:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

► **Rank 2 ( $\mathbf{v}_2$ ):** Solving  $(B - I) \mathbf{v}_2 = \mathbf{v}_1$ :

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

The system requires  $g_2 = 1$ . Choosing the free variable  $g_1 = 0$ , we get the **generalized eigenvector of rank 2** :

$$\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Observe that the set  $\{v_1, v_2\}$  is a basis for  $\mathbb{R}^2$ .

**Verification:** The rank 2 condition is  $(B - I)^2 \mathbf{v}_2 = \mathbf{0}$  but  $(B - I) \mathbf{v}_2 = \mathbf{v}_1 \neq \mathbf{0}$ .

$$(B - I) \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \mathbf{0} \quad \text{and} \quad (B - I)^2 \mathbf{v}_2 = (B - I) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

**Example (Generalized Eigenvector of Rank 3).** Consider the matrix

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Observe that all the eigenvalues of  $A$  are 2, simply because it is upper triangular. The operator  $A - 2I$  is:

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Note that  $(A - 2I) \mathbf{e}_4 = \mathbf{0}$ , and the equation  $(A - 2I) \mathbf{v} = \mathbf{e}_4$  has not solution. However, we have :

$$\begin{aligned} (A - 2I) \mathbf{v}_3 &= \mathbf{v}_2 \neq \mathbf{0} \\ (A - 2I)^2 \mathbf{v}_3 &= (A - 2I) \mathbf{v}_2 = \mathbf{v}_1 \neq \mathbf{0} \\ (A - 2I)^3 \mathbf{v}_3 &= (A - 2I) \mathbf{v}_1 = \mathbf{0} \end{aligned}$$

Thus,  $\mathbf{v}_3 = (0, 0, 1, 0)^\top$  is a *generalized eigenvector of rank 3*. It is the crucial starting vector that generates the entire 3-dimensional invariant subspace whose basis is  $\{v_1, v_2, v_3\}$

**Chain Equations:** The chain is defined by:

1.  $(A - 2I) \mathbf{v}_1 = \mathbf{0}$  (Eigenvector, Rank 1)
2.  $(A - 2I) \mathbf{v}_2 = \mathbf{v}_1$  (Generalized Eigenvector, Rank 2)
3.  $(A - 2I) \mathbf{v}_3 = \mathbf{v}_2$  (Generalized Eigenvector, Rank 3)

The Jordan Canonical Form (JCF) provides the "next best" form for a matrix that is not diagonalizable. It is the closest we can get to a diagonal matrix through similarity transformations ( $A \sim J = P^{-1}AP$ ). The basis for a Jordan block of size  $k \times k$  is given by the  $k$  vectors in its corresponding Jordan chain. Jordan canonical form is built upon Jordan blocks which is defined as :

**Definition (Jordan Block).** A **Jordan block**  $J_k(\lambda)$  is a  $k \times k$  upper triangular matrix with the eigenvalue  $\lambda$  on the main diagonal and 1's on the superdiagonal:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & 0 & 0 & \lambda \end{pmatrix}$$

The number of Jordan blocks corresponding to an eigenvalue  $\lambda$  is equal to its Geometric Multiplicity ( $\mathbf{GM}(\lambda)$ ).

**Example** (Matrix with Two Jordan Blocks:  $4 \times 4$ ). Consider a  $4 \times 4$  matrix  $A$  with a single eigenvalue  $\lambda = 2$ , where the total algebraic multiplicity  $\mathbf{AM}(2) = 4$  and the geometric multiplicity  $\mathbf{GM}(2) = 2$ .

Since  $\mathbf{GM} = 2$ , the Jordan form  $J$  must have two separate Jordan blocks. The size of the blocks must sum to  $\mathbf{AM} = 4$ . The only partition is  $3 + 1$ .

The Jordan Canonical Form is:

$$J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} J_3(2) & \mathbf{0} \\ \mathbf{0} & J_1(2) \end{pmatrix}$$

A matrix  $A$  that is similar to this  $J$  (i.e.,  $A = P^{-1}JP$  for some invertible matrix  $P$ ) has a basis of generalized eigenvectors structured as:

► **Chain 1 (Length 3):**  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ . Here,  $\mathbf{v}_1$  is the eigenvector,  $\mathbf{v}_2$  is a generalized eigenvector of rank 2, and  $\mathbf{v}_3$  is a generalized eigenvector of rank 3.

► **Chain 2 (Length 1):**  $\{\mathbf{v}_4\}$ . Here,  $\mathbf{v}_4$  is a second, linearly independent eigenvector.

**Example.** Consider a  $5 \times 5$  matrix  $B$  with a single eigenvalue  $\lambda = -1$ , where  $\mathbf{AM}(-1) = 5$  and the geometric multiplicity  $\mathbf{GM}(-1) = 3$ .

Since  $\mathbf{GM} = 3$ , the Jordan form  $J$  must have three separate Jordan blocks. The sizes must sum to  $\mathbf{AM} = 5$ . Possible partitions are  $3 + 1 + 1$  or  $2 + 2 + 1$ . We choose  $3 + 1 + 1$ .

The Jordan Canonical Form is:

$$J = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} J_3(-1) & & \mathbf{0} \\ & J_1(-1) & \\ \mathbf{0} & & J_1(-1) \end{pmatrix}$$

A matrix  $B$  similar to this  $J$  (e.g.,  $B = J$ ) has a basis of generalized eigenvectors structured as:

► **Chain 1 (Length 3):**  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ .

► **Chain 2 (Length 1):**  $\{\mathbf{v}_4\}$ .

► **Chain 3 (Length 1):**  $\{\mathbf{v}_5\}$ .

**Interpretation:** The matrix  $B$  has three distinct invariant subspaces (one 3D and two 1D) that are spanned by these Jordan chains. Within the 3D subspace, the transformation acts as a combination of scaling (by  $\lambda = -1$ ) and a linear shear (due to the 1's on the superdiagonal).

**Definition** (Jordan Canonical Form (JCF)). A matrix  $J$  is in **Jordan Canonical Form** if it is a block diagonal matrix whose blocks are Jordan blocks:

$$J = \begin{pmatrix} J_{k_1}(\lambda_1) & & & 0 \\ & J_{k_2}(\lambda_2) & & \\ & & \ddots & \\ 0 & & & J_{k_m}(\lambda_m) \end{pmatrix}$$

where the  $\lambda_i$ 's are the eigenvalues of  $A$  (not necessarily distinct).

**Example** (Generalized Eigenvectors of  $J_3(\lambda)$ ). We aim to prove that the generalized eigenvectors of the canonical  $3 \times 3$  Jordan block  $J$  are the standard basis vectors  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ .

Let  $J$  be the  $3 \times 3$  Jordan block for an eigenvalue  $\lambda$ :

$$J = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

The operator defining the Jordan chain is the nilpotent matrix  $N = J - \lambda I$ :

$$N = J - \lambda I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The Jordan chain of length 3 is defined by the three relations, where  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  are the generalized eigenvectors:

$$N\mathbf{v}_3 = \mathbf{v}_2 \quad (\text{Rank 3})$$

$$N\mathbf{v}_2 = \mathbf{v}_1 \quad (\text{Rank 2})$$

$$N\mathbf{v}_1 = \mathbf{0} \quad (\text{Rank 1, Eigenvector})$$

We show that the standard basis vectors,  $\{\mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_1\}$ , satisfy this chain in reverse order (i.e.,  $\mathbf{v}_3 = \mathbf{e}_3$ ,  $\mathbf{v}_2 = \mathbf{e}_2$ ,  $\mathbf{v}_1 = \mathbf{e}_1$ ).

- Eigenvector  $\mathbf{v}_1$  (Rank 1) :

We test  $\mathbf{e}_1$ :

$$N\mathbf{e}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{0}$$

The vector  $\mathbf{e}_1$  is indeed the eigenvector,  $\mathbf{v}_1$ .

- Generalized Eigenvector  $\mathbf{v}_2$  (Rank 2)

We test  $\mathbf{e}_2$ :

$$N\mathbf{e}_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \mathbf{e}_1$$

The vector  $\mathbf{e}_2$  satisfies  $N\mathbf{e}_2 = \mathbf{e}_1 = \mathbf{v}_1$ . It is the generalized eigenvector of rank 2,  $\mathbf{v}_2$ .

- Generalized Eigenvector  $\mathbf{v}_3$  (Rank 3)

We test  $\mathbf{e}_3$ :

$$N\mathbf{e}_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \mathbf{e}_2$$

The vector  $\mathbf{e}_3$  satisfies  $N\mathbf{e}_3 = \mathbf{e}_2 = \mathbf{v}_2$ . It is the generalized eigenvector of rank 3,  $\mathbf{v}_3$ .

## Conclusion

The standard basis vectors satisfy the defining equations of the Jordan chain:

$$N\mathbf{e}_3 = \mathbf{e}_2, \quad N\mathbf{e}_2 = \mathbf{e}_1, \quad N\mathbf{e}_1 = \mathbf{0}$$

Therefore, for the **canonical** Jordan block  $J_3(\lambda)$ , the generalized eigenvectors are the standard basis vectors:  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ .

**Theorem 2.1** (Jordan Canonical Form). *Let  $A$  be an  $n \times n$  matrix whose characteristic polynomial splits over the field (e.g., any matrix over  $\mathbb{C}$ , or a matrix over  $\mathbb{R}$  whose eigenvalues are all real). Then  $A$  is similar to a matrix  $J$  in Jordan Canonical Form:*

$$A = PJP^{-1}$$

*The matrix  $J$  is unique up to the ordering of the Jordan blocks. The columns of  $P$  form a basis of **generalized eigenvectors**.*

## Key Connection to Multiplicity

The structure of the Jordan blocks reveals exactly why diagonalization that we discussed fails:

1. **Size of  $J$ :** The sum of the sizes of all Jordan blocks ( $k_1 + k_2 + \dots + k_m$ ) must equal the dimension  $n$ .
  2. **Algebraic Multiplicity (AM):** The  $\mathbf{AM}(\lambda)$  is the total number of times  $\lambda$  appears on the diagonal of  $J$ .
  3. **Geometric Multiplicity (GM):** The  $\mathbf{GM}(\lambda)$  is the number of distinct Jordan blocks associated with  $\lambda$ .
- **Diagonalizable Case:**  $\mathbf{GM}(\lambda) = \mathbf{AM}(\lambda)$ . Every Jordan block is  $1 \times 1$ :  $J = \text{diag}(\lambda_1, \lambda_2, \dots)$ .
  - **Non-Diagonalizable Case:**  $\mathbf{GM}(\lambda) < \mathbf{AM}(\lambda)$ . There must be at least one Jordan block of size  $k \geq 2$ . The difference  $\mathbf{AM}(\lambda) - \mathbf{GM}(\lambda)$  is related to the maximum size of the Jordan blocks.

**Example** (Jordan Form for a  $2 \times 2$  Non-Diagonalizable Matrix). Consider the matrix  $B$  from the recap, which failed to diagonalize:

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

- Eigenvalues and Multiplicities:

The only eigenvalue is  $\lambda = 1$ , with  $\mathbf{AM}(1) = 2$ . The Geometric Multiplicity is  $\mathbf{GM}(1) = \dim(\ker(B - I)) = \dim\left(\ker\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) = 1$ . Since  $\mathbf{GM} < \mathbf{AM}$ ,  $B$  is not diagonalizable, and the JCF must contain only **one** Jordan block (since  $\mathbf{GM} = 1$ ) of size  $2 \times 2$  (since  $\mathbf{AM} = 2$ ).

- The Jordan Form and Generalized Eigenvectors:  
The Jordan Canonical Form is simply the  $2 \times 2$  Jordan block:

$$J = J_2(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Since  $B = J$ ,  $P = I$ . The single eigenvector is  $v_1 = (1, 0)^\top$ .

The second column of the basis  $P$  is a **generalized eigenvector**  $g$  that forms a Jordan chain:

$$(B - \lambda I)v_1 = 0 \quad (\text{Eigenvector})$$

$$(B - \lambda I)g = v_1 \quad (\text{Generalized Eigenvector})$$

For  $B$  and  $\lambda = 1$ :

$$(B - I)g = v_1 \implies \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

This gives  $g_2 = 1$  (and  $g_1$  is free). Choosing  $g_1 = 0$ , we get  $g = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . The transformation  $B$  acts on this Jordan basis  $\{v_1, g\}$  as:  $Bv_1 = v_1$  and  $Bg = v_1 + g$ . The off-diagonal 1 in the JCF represents this "shear" component.

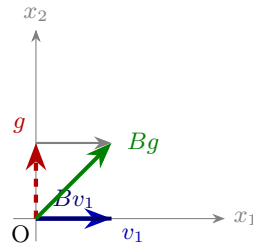


Figure 2: Action of  $B$  on the Jordan basis  $\{v_1, g\}$ .  $v_1$  is an invariant direction ( $Bv_1 = v_1$ ), while  $g$  is 'sheared' into the direction of  $v_1$  ( $Bg = v_1 + g$ ).

**Example.** The Jordan Canonical Form of a matrix  $A$  does not exist if the base field over which the matrix is defined is not **algebraically closed**, and consequently, the characteristic polynomial of  $A$  does not split completely (factor entirely into linear terms) over that field. The most common example is a matrix with real entries. Consider :

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

- Characteristic Polynomial : The characteristic polynomial  $p(\lambda)$  is given by  $\det(A - \lambda I)$

$$p(\lambda) = \det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} = (-\lambda)(-\lambda) - (-1)(1) = \lambda^2 + 1$$

- Eigenvalues : Setting  $p(\lambda) = 0$  to find the eigenvalues:

$$\lambda^2 + 1 = 0 \implies \lambda^2 = -1 \implies \lambda = \pm i$$

We see that over  $\mathbb{R}$  the matrix  $A$  does not have any eigenvalues. So, the matrix does not admit a Jordan canonical form. On the other hand, over  $\mathbb{C}$ ,  $A$  may be diagonalized - how? - and has the following diagonal form

$$J = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

It should be noted that a matrix being defined over an algebraically closed field - such as  $\mathbb{C}$  - does not imply that it is diagonalizable. While working over an algebraically closed field guarantees that all eigenvalues exist and the **Jordan Canonical Form (JCF) exists**, it does not guarantee that the matrix can be diagonalized. The following example explains this :

**Example.** Consider the  $2 \times 2$  matrix  $A$  defined over  $\mathbb{C}$ :

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

- Eigenvalues and Algebraic Multiplicity (AM) : The characteristic polynomial  $p(\lambda)$  is:

$$p(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 0 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2$$

The only eigenvalue is  $\lambda = 2$ . The **Algebraic Multiplicity** is:

$$\text{AM}(2) = 2$$



- **Geometric Multiplicity (GM)** : The geometric multiplicity is the dimension of the null space of  $(A - 2I)$ .

$$A - 2I = \begin{pmatrix} 2-2 & 1 \\ 0 & 2-2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

We solve the system  $(A - 2I)x = 0$ , which yields the equation  $0x_1 + 1x_2 = 0$ , or  $x_2 = 0$ . The eigenvectors  $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$  are of the form:

$$x = \begin{pmatrix} x_1 \\ 0 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_1 \in \mathbb{C}, x_1 \neq 0$$

The eigenspace is spanned by the single vector  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ . The **Geometric Multiplicity** is:

$$\text{GM}(2) = 1$$

Since  $\text{AM}(2) = 2$  and  $\text{GM}(2) = 1$ , we have  $\text{AM}(2) \neq \text{GM}(2)$ . Therefore, the matrix  $A$  is **not diagonalizable**.  $A$  is already in its Jordan Canonical Form (a single Jordan block).

The Schur Decomposition is a foundational result in matrix theory. Unlike the Jordan Canonical Form, which only exists when the characteristic polynomial splits over the base field, the Schur Decomposition always exists for any square matrix over the complex numbers  $\mathbb{C}$  providing a unitarily similar triangular form. In numerical linear algebra and machine learning, the *Schur decomposition* provides a stable alternative to the Jordan canonical form. It guarantees the existence of an orthogonal (or unitary) basis in which any matrix becomes triangular, while preserving the spectrum. This decomposition underlies modern eigenvalue algorithms (such as the QR iteration) and spectral methods in machine learning.

## 2.1 Complex Schur Decomposition

**Definition** (Hermitian Adjoint and Unitary Matrix). Let  $A \in \mathbb{C}^{m \times n}$  be a complex matrix. The **Hermitian adjoint** (or **conjugate transpose**) of  $A$ , denoted by  $A^*$ , is defined as

$$A^* = \overline{A}^T,$$

that is, the transpose of  $A$  with each entry replaced by its complex conjugate.

A square matrix  $U \in \mathbb{C}^{n \times n}$  is called **unitary** if

$$U^*U = UU^* = I_n.$$

Equivalently, the columns (and rows) of a unitary matrix form an **orthonormal basis** of  $\mathbb{C}^n$  with respect to the standard inner product

$$\langle x, y \rangle = y^* x.$$

**Example** (Real Case and Geometric Interpretation). If  $U$  is real,  $U^* = U^T$ , and the unitarity condition  $U^T U = I$  means that  $U$  is an **orthogonal matrix**. Orthogonal (or unitary) matrices represent rotations and reflections that preserve inner products, lengths, and angles. For instance,

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

is orthogonal since  $U^T U = I$ . It rotates vectors in  $\mathbb{R}^2$  by  $45^\circ$  while preserving their Euclidean norm.

**Definition** (Schur Decomposition over  $\mathbb{C}$ ). Let  $A \in \mathbb{C}^{n \times n}$ . There exists a **unitary** matrix  $Q$  (so that  $Q^*Q = I$ ) and an **upper triangular** matrix  $T$  such that

$$A = QTQ^*.$$

The matrix  $T$  is called the **Schur form** of  $A$ , and the diagonal entries of  $T$  are the eigenvalues of  $A$ .

**Example** (Triangularization of a Complex Matrix). Let

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}.$$

The eigenvalues are  $\lambda_{1,2} = 2.5 \pm 1.3229i$ . Hence there exists a unitary matrix  $Q$  such that

$$T = Q^* A Q = \begin{bmatrix} 2.5 + 1.3229i & * \\ 0 & 2.5 - 1.3229i \end{bmatrix}.$$

Although  $A$  is not diagonalizable over  $\mathbb{R}$ , its Schur form  $T$  is upper triangular, with the eigenvalues on the diagonal. This representation is numerically stable and preferred for computations.